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Why This Project: The Reference Disappears

Conventional: the machine sets the reference



$f, \angle V$ held by rotating mass

Islanded: nothing defines the angle



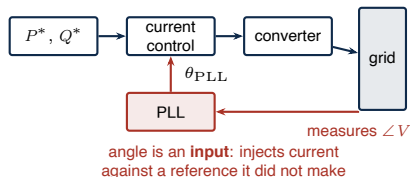
every GFL converter loses its reference

The problem

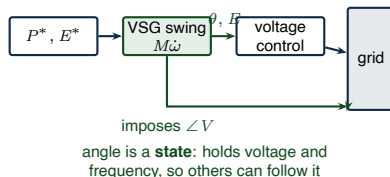
As synchronous generation is displaced by converters, the network keeps the power but loses the *source of its voltage angle and frequency*. A grid-following converter cannot supply what it is built to measure, so an island of GFL units has no reference at all — and a converter left permanently grid-forming gives up the tracking accuracy it was dispatched for.

What GFL and GFM Actually Mean

Grid-following (GFL)



Grid-forming (GFM)



The consequence for this project

Neither mode is better in general. GFL tracks its dispatch accurately *while a reference exists*; GFM supplies that reference but carries the island's frequency and voltage on its own swing state. The engineering question is therefore not which mode to pick, but **which converters must form, when, and how many**.

What This Project Provides

Fixed choice, decided offline

- mode set once, by hand
- forming count guessed in advance
- no evidence it is admissible
- cannot answer a new disturbance



Decided from the network state

- severity index drives the transition
- admissible GFM sets screened by SSSA
- each transition authenticated first
- refuses rather than guesses

- One in-house chain — **Power Flow** → **SSSA** → **Time-Domain Simulation** — on a single shared model, so a screening result and a simulated trajectory describe the same system.
- Four converters with one common plant and **selectable GFL/GFM controller branches**, switched at run time with current and voltage continuity.
- The number of forming units is an **output** of the network condition, not an assumption.
- Verified across mode transitions, island support, and SG reconnection under an event-driven disturbance chronology.